



Extrapolating a sequence

1 Which of these could be a rule for generating the sequence 8, 16, 32, ...? Tick **2** correct answers

- ☐ Multiply the previous term by 2 to get the next term.
- ☐ Add 8 to the previous term to get the next term.
- ☐ Add 8, then add 16 then add 24; common second difference of 8.
- ☐ Add the two previous terms to get the next term.

2 Izzy says the next term in the sequence 1, 3, 6, 10, ... must be 15. Why is she incorrect? Tick **1** correct answer

- ☐ These are a triangular numbers, they are not a sequence.
- ☐ There is not a common difference of 5 between successive terms.
- ☐ The pattern may change after the fourth term.
- ☐ Sequences must have a constant additive or multiplicative rule.

3 Which of these could be a rule for the sequence: 37, 41, 43, 47, 53, ...? Tick **1** correct answer

- ☐ Prime numbers over 30 written in ascending order
- ☐ A sequence which alternates between adding 4 and adding 2
- ☐ A linear sequence with common difference +4
- ☐ Sequence of numbers one more than the square numbers
- ☐ Integers where the digits sum to multiples of 5

4 Which of these could be the rule for the sequence starting 17, 19, 20, ...? Tick **2** correct answers

- ☐ Odd numbers greater than 15.
- ☐ Add 2, add 1, add 0, add -1; common second difference of -1.
- ☐ Integers greater than 16 where the digits sum to even numbers.
- ☐ Numbers with nine letters when written in English.
- ☐ Add the two previous numbers and subtract 10 to get the next number.



5 Which of these could be the rule for the sequence starting 6, 7, 9, ...? Tick **3** correct answers

- ☐ Linear sequence with a common difference of +2
- ☐ Sequence which adds 1, then 2, then 3; common second difference of 1
- ☐ Double the previous term then subtract 5 to get the next term
- ☐ Number of letters in the English words for days of the week, starting on Monday
- ☐ Number of letters in the English words for the months starting on August

6 Match each sequence which start with the terms 3, 6, ... to the correct rule. Write the correct letter in each box

a	3, 6, 9, 12, ...
b	3, 6, 12, 24, ...
c	3, 6, 10, 15, ...
d	3, 6, 12, 21, ...
e	3, 6, 15, 42, ...

	triangular numbers greater than 1.
	add 3, add 6, add 9, common second difference of 3.
	triple the previous term and subtract 3 to get the next term.
	add 3 to the previous term to get the next term.
	multiply the previous term by 2 to get the next term.

